AegisHealth AI — Phase 1 MVP Architectural Guide

Healthcare Hackathon Foundation • System Architecture, Workflow & Feature Breakdown

Executive Summary: AegisHealth AI is a clinical intelligence workstation designed to prevent the $40B annual crisis of Adverse Drug Events (ADEs) and polypharmacy contraindications. Built in strict compliance with the 24-Hour Healthcare Hackathon Blueprint, it features an Adversarial Multi-Agent Debate Protocol (Clinical Drafter vs. Safety Auditor), Live Glassbox Thought-Chain Stepper, Dual-Persona View (Clinician SOAP vs. Patient 5th-Grade Plain English), and HL7 FHIR R4 EHR Interoperability.

1. Clean Workspace & Folder Structure

The project follows Next.js 15 App Router conventions with a strict separation of concerns:

```
utkarsh/
■■■ docs/
■ ■■■ PRD.md # 10-section enterprise clinical requirements
■ ■■■ DESIGN.md # Clinical design tokens, typography, and anti-AI guidelines
■ ■■■ TECHSTACK.md # API contracts, dependency matrix, and data flow
■ ■■■ schema.sql # PostgreSQL / Supabase DDL with Row Level Security
■■■ src/
■ ■■■ app/ # Next.js App Router (page.tsx, layout.tsx, globals.css)
■ ■ ■■■ api/ # /api/pipeline (Groq debate) & /api/audit (logging)
■ ■■■ components/ # 9 modular UI components (Dashboard, Pills, Stepper, etc.)
■ ■■■ data/ # mockData.ts (3 rich scenarios: ER, Conflict, Discharge)
■ ■■■ lib/ # groq.ts, supabase.ts, fhir.ts, auth.ts, utils.ts
■■■ package.json # Next 15.2.1, React 19, Lucide, Framer Motion, Groq SDK
■■■ README.md # Complete quickstart, architecture, and pitch guide
```

2. End-to-End Clinical Data Workflow

The application coordinates a 5-stage glassbox data pipeline from ingestion to EHR push:

Stage	Pipeline Action	Safeguard / Technology
1. Ingestion & Consent	Input via text, simulated OCR, or voice; check ABDM consent token.	Blocks execution with HTTP 403 if patient revoked consent.
2. Multi-Agent Debate	Agent A (Drafter) proposes SOAP note; Agent B (Auditor) verifies safety.	Groq Llama 3.3 70B sub-500ms token generation + offline fallba
3. Live Thought-Chain	Glassbox Stepper streams: [1] PHI Redacted -> [2] RxNorm -> [3] Audit.	Framer Motion animated progression; zero opaque spinners.
4. Dual-Persona Canvas	Renders Clinician SOAP (ICD-10) OR Patient 5th-grade plain language.	Includes Attending Doctor Digital Attestation Sign-Off Gate.
5. FHIR EHR Export	Serializes into compliant HL7 FHIR R4 Bundle with JSON download.	DISHA / ABDM & HIPAA append-only audit trail in Supabase.

3. Deep-Dive Feature Breakdown (UI & Implementation)

Feature 1: Universal Split-Screen Layout — Component: SplitScreenDashboard.tsx

Responsive 50/50 dual-pane architecture. The Left Pane hosts multi-modal inputs, scenario quick pills, and patient vitals. The Right Pane streams real-time reasoning and tabbed clinical/patient outputs. Sticky header maintains branding, RBAC selector, latency telemetry, and FHIR export trigger.

Feature 2: 1-Click Preset Scenario Buttons — Component: ScenarioPills.tsx & mockData.ts

Preloads Scenario A (ER Triage & Vitals: Aditi Rao, 47F), Scenario B (Polypharmacy Conflict: Vikram Malhotra, Warfarin + Ibuprofen), and Scenario C (Discharge Summary: Ramesh Gupta, COPD + steroid taper). Enables a polished, zero-typing 90-second judge walkthrough.

Feature 3: The Multi-Agent Debate Protocol — Component: groq.ts & /api/pipeline/route.ts

Implements adversarial verification. Agent A (Clinical Drafter) writes initial notes; Agent B (Safety Auditor) evaluates contraindications. If unsafe, emits a CRITIQUE and flags risk score (82%). Only responds 'APPROVED' when safe, preventing model hallucinations.

Feature 4: Live Glassbox Thought-Chain Stepper — Component: ThoughtChain.tsx

Replaces generic spinners with animated execution milestones ([1] PHI Redaction, [2] RxNorm, [3] Safety Audit, [4] Persona Synthesis) showing live millisecond latencies and risk badges.

Feature 5: Clinical Contraindication Alert System — Component: ClinicalAlerts.tsx

Strictly uses the reserved alert color #EF4444 to highlight critical drug-drug conflicts. Displays toxicity risk score (82%), biochemical mechanism (COX-1 displacement), and required counteractions (Paracetamol substitution, INR check).

Feature 6: Dual-Persona View & Regional Localization — Component: DualPersonaView.tsx

Instant switch between Clinician View (dense SOAP note, ICD-10 codes, RxNorm IDs) and Patient View (Flesch-Kincaid 5th-grade plain English, visual pill schedule, and Hindi/Telugu translations).

Feature 7: Human-in-the-Loop Attestation Gate — Component: DualPersonaView.tsx

Enforces Healthcare Safety Principles: AI outputs remain 'Pending MD Attestation' until an attending physician clicks 'Digitally Sign Off & Approve', stamping the note with license credentials before EHR dissemination.

Feature 8: DISHA / ABDM Patient Consent Framework — Component: ConsentBanner.tsx

Displays active consent token ABDM-CONSENT-9942. When logged in as a patient, allows 1-click consent revocation, which immediately halts downstream AI execution with HTTP 403.

Feature 9: Role-Based Access Control (RBAC) — Component: AuthRoleSelector.tsx & auth.ts

Provides 1-click persona switching between Dr. Priya Sharma, MD (Attending Physician), Sunita Nair, RN (Nurse Coordinator), and Rajesh Patel (Patient) with tailored permissions and restricted views.

Feature 10: HL7 FHIR R4 Interoperability Modal — Component: FhirExportModal.tsx & fhir.ts

Generates compliant FHIR R4 Bundle JSON with Patient, Observation, Condition, and MedicationRequest resources, featuring one-click copy and .json file download for EHR integration.

Feature 11: Backend API & Database Architecture — Component: /api/pipeline, /api/audit & docs/schema.sql

Server-side Next.js route handlers with Groq Llama 3.3 70B integration, and complete PostgreSQL schema with Row-Level Security (RLS) across patients, encounters, debates, and audits.

4. The Winning 3-Minute / 90-Second Demo Script

Structured according to Chapter 7 of the Blueprint for maximum hackathon scoring:

Timing	Demo Phase	On-Screen Action & Spoken Script
0:00 - 0:20	Hook	Quote: 'Every year, over 5 lakh patients in India suffer preventable adverse drug events from polypharmacy c
0:20 - 0:40	Problem	Quantify: 'Meet Vikram Malhotra, a 68-year-old on warfarin who was prescribed ibuprofen for joint pain—a 4x
0:40 - 1:10	Live Demo (Debate)	Click 'Scenario B: Drug Conflict' -> Click 'Run Pipeline' -> Show Thought-Chain step [3] flag 82% risk -> Show
1:10 - 1:30	Attestation & Persona	Show Doctor Sign-Off -> Switch to Patient Rajesh Patel -> Show view flip to 5th-grade language + Hindi/Telu
1:30 - 2:00	EHR & Architecture	Show ABDM Consent revocation -> Click 'Export to EHR (FHIR)' to download JSON. Close: 'Sub-500ms Gro

5. Quality Assurance & Build Verification

- **TypeScript Compilation (tsc --noEmit):** 0 errors across all types, interfaces, and API routes.
- **Next.js Production Build:** Successfully compiled and prerendered all 7 static and dynamic API routes.
- **Zero-Delay Fallback:** Tested and verified to run 100% offline if Wi-Fi drops, preventing demo crashes.